

10. (3 points)

a. (0.5 point)

Define external systemic risk and internal systemic risk.

b. (1 point)

Identify and briefly describe two types of external systemic risk.

c. (1 point)

Identify and briefly describe two types of internal systemic risk.

d. (0.5 point)

Describe how the choice of valuation classes within a claims portfolio can affect internal systemic risk.

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QUESTION 10	
TOTAL POINT VALUE: 3	LEARNING OBJECTIVE: A5: Describe the various sources of risk and uncertainty that are associated with the determination of reserves. Calculate risk margins that consider these sources of risk and uncertainty.
SAMPLE ANSWERS	
Part a: 0.5 point	
<i>Sample Answer 1</i> External systemic risks are risks external to the insurance modeling process that impact all valuation classes and claim groups. Internal systemic risks are risks that are internal to the insurance modeling process that impact all valuation classes and claim groups. <i>Sample Answer 2</i> Internal systemic risk – Risk internal to insurance liability modeling/valuation. It represents risk arising due to the fact that model can't fully replicate insurance process. External Systemic Risk – Risk external to the insurance liability valuation/modeling process. It arises due to external changes in environment having an impact on insurance liabilities/models.	
Part b: 1 point	
<i>Sample Answer 1</i> Event risk- Risk from catastrophes/disasters either natural or manmade. Economic/Social risk-risk from economic/social changes like inflation. <i>Sample Answer 2</i> Event risk- Risk from large unpredicted event causing many losses to insurer (e.g. Cat). Recovery risk- Risk from recovery (reinsurance and non-reins) in salvage and subro / reinsurer ability to pay claims.	
Part c: 1 point	
<i>Sample Answer 1</i> Parameter selection error – uncertainty that the model will not be able to capture all the parameters and the trends. Data error-uncertainty due to a lack of credible data, or lack of knowledge about the data. <i>Sample Answer 2</i> Data error – risk arising from lack of credible data, inadequate understanding of portfolio analyzed. Parameter selection error – arises from fact that model can't adequately measure all	

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predictors of future claims costs or trends in those predictors.
Part d: 0.5 point
<i>Sample Answer 1</i> Internal systemic risk may be impacted by the size of the valuation classes. If there are many valuation classes that are small in size, internal systemic risk will increase because the data will be more volatile. Internal systemic risk could also increase if non-homogeneous claims grouped are placed in the same valuation class.
<i>Sample Answer 2</i> Choice must ensure valuation classes grouped together are mostly homogenous and in line with central estimate valuation. To reduce risk (of selection in parameters and others), they must show similar qualifications (e.g., mostly same development pattern) and must be sufficiently large data sets (don't split too much) so that data is still credible and reliable.
EXAMINER'S REPORT
Part a
Candidates were expected to define internal and external systemic risk. The key point to earn full credit was "internal/external to the actuarial/valuation/reserving/modeling/estimation process". A large number of candidates did not earn full credit. Common errors included: • Internal/external to the company/insurer • Within/outside of the company's control • Diversifiable or non-diversifiable • Inside/outside of underwriting process or insurance process (no credit was given because underwriting or insurance process is too vague unless there is further explanation in the response) • Only listing a few examples, such as "external systematic risk includes economic and social risk, event risks, etc.", or responses like "internal risk related to / associated with the valuation process, such as parameter selection error or data error" • Internal risk is risk around / deals with valuation process (with no further explanation) • External = everything else
Part b
Candidates were expected to identify and properly describe two external systemic risks. Most candidates did well on this part. Common errors included: • Simply listing an example, such as Event Risk: e.g. CAT (Note: the only exception is economic and social risk; even if the candidate simply listed inflation, credit was given for description because there is not much explanation in the text book for this particular external risk, but for all other external system risks, proper descriptions were required for full credit.) • Mismatch between identification and description (e.g., listing economic risk, but

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providing a description of the claim handling process).
Part c
Candidates are expected to identify and properly describe two internal systemic risks. Most candidates did well on this part. Common errors include: • Mismatch between identification and description (e.g., listing specification error, but providing a description of parameter selection error). • Internal fraud • The same actuary effect Note: A few candidates mixed their responses for parts (b) and (c).
Part d
Candidates were expected to recognize the benefit of grouping valuation classes properly considering the homogeneity, credibility, and/or operation efficiency. For full credit, the candidate had to articulate the benefits of both more homogeneous and more heterogeneous valuation class groupings. Candidates frequently misunderstood the question. Common errors included: • no answer, • no coherent answer, • responses talking about how class selections need to avoid the same actuary effect, • statements about how the incorrect selection of valuation classes can lead to unwanted correlations between classes, without any discussion of the issue of valuation class homogeneity versus data credibility issues.